Tips for Preservation

In vegetables, enzyme action may produce undesirable effects on flavour and texture during freezing. So the enzymes must be destroyed by heating before the vegetables are frozen.

Freezing may *require three to seventy-two hours* under such conditions. *Home freezing* is done by this method.

- **Quick freezing process**

The temperatures used in the *quick freezing process* range from *–32°C to –40°C.* It freezes food so rapidly that fine crystals are formed. The time taken for quick freezing is significantly lower than that of slow freezing. In quick freezing, *large quantities of food* can be frozen in a short period of time. The use of very low temperature for both freezing and holding frozen products adds to the cost but of *desirable for* many products in terms of *retention of palatability* and *nutritive value.*

- **Dehydro freezing**

Dehydro freezing of fruits and vegetables is the drying of the food to about 50 percent of its original weight and volume and then freezing the food to preserve it. The quality of dehydro frozen fruits and vegetables is equal to that of fruits and vegetables which are frozen without preliminary drying. The cost is marginally less because of weight and volume savings in packing, freezing, storing and shipping.

DO YOU KNOW?

Cryogenic liquids are liquefied gases used for freezing due to its low boiling point. The commonly used cryogenic liquids are liquid nitrogen, liquid carbon-dioxide and Freon.

Points to be Considered Before Freezing Food

Vegetables:

Blanching (dipping the products in boiling water for two to three minutes) vegetables before freezing reduces the number of microorganisms, removes some air from the tissues, makes them more compact and enhances their colour. Its most important function is to *inactivate enzymes* otherwise that would cause deterioration in palatability, colour and ascorbic acid content during storage.

Fruits:

The enzymes of fruits can be inactivated by *blanching* but it *is not done* as it gives the *fruit a cooked flavour and soft texture*. Rather fruits are cut directly into sugar syrup or sugar to prevent oxidation. Sugar not only increases the sweetness but helps to retain volatile aroma.

Meat and poultry:

Meat and poultry require only *wrapping for freezing*. After slaughtering the animal, the pork, meat and poultry is chilled promptly to avoid spoilage. The tendency of the fat of the pork and poultry to become rancid during storage in a freezer is aggravated by storage before freezing.

4.2.2 Preservation by High Temperature

The temperature and time used in heat processing a food depends upon the effects of heat on food and the other preservative methods employed.

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